Simulated Annealing Bisection: Cooling the Edge-Cut Objective in the Redistricting Tree

U.3 — Algorithm Design / Structure Layer

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Abstract

We introduce SimulatedAnnealing bisection, a new Structure-layer algorithm for the redistricting tree in which each node is solved not by a single Metis call but by a simulated annealing (SA) loop seeded from the Metis solution. SA accepts boundary-tract flips according to a geometric temperature schedule, allowing exploration of worse solutions early and converging to a local optimum at the end. The cooling schedule is scaled to the subgraph size—$n_{\text{steps}} = \texttt{steps\_per\_tract} \times m$ at a node with m tracts—so that leaf nodes and root nodes receive proportional computational effort. Crucially, the algorithm tracks the *best-ever* plan seen, not the final state after cooling; this discards the exploration overhead and returns the lowest edge-cut bisection found during the run.

Empirically, SimulatedAnnealing with $\texttt{steps\_per\_tract} = 10$ achieves 10–18% Polsby-Popper improvement over baseline Metis bisection on North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$). This improvement lies between multi-seed Metis and BisectionEnsemble at $T = 100$, at a runtime cost that scales as $O(\texttt{steps\_per\_tract} \times n)$ per node. SimulatedAnnealing is a deterministic optimiser:

it produces a single high-quality bisection per node rather than a distributional sample. We contrast it with ReCom (Markov chain sampling) and BisectionEnsemble (probabilistic seed search), and identify when each is preferred.

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1. Introduction

1.1. The Local-Optimum Problem in METIS Bisection

The core computational primitive of the B-series redistricting platform is the bisection node: given a subgraph of m census tracts that must be divided into two contiguous, population-balanced pieces, find the cut that minimises the weighted edge cut [Della-Libera, 2026e,a]. The standard solver is METIS, a multilevel graph partitioner that coarsens the graph, finds an approximate minimum bisection on the coarsened graph, and uncoarsens with Kernighan-Lin (KL) local refinement [Karypis and Kumar, 1998]. METIS is fast—$O(m \log m)$ per call—and produces solutions within a small constant factor of optimal on typical redistricting graphs.

However, METIS is a heuristic. Graph minimum bisection is $\mathcal{NP}$-hard [Garey et al., 1976], and the KL refinement phase terminates at a local optimum that depends on the random seed. The B.7 analysis showed that for most states, seed variation has little practical effect on the final map, but the variation is nonzero and occasionally significant [Della-Libera, 2026d]. The B-series response to this fragility has followed two tracks.

Track 1: Better seed search. CONVERGENCESWEEP (U.1) sweeps forward through seeds, running METIS at each, and halts when $T = 600$ consecutive seeds produce no improvement in normalised edge cut [Della-Libera, 2026b]. BISECTIONENSEMBLE (U.9) runs T independent METIS seeds in parallel and selects the minimum [Della-Libera, 2026i]. Both are *SeedCompositor* strategies—they change how many times the structure-layer bisector is called, not how it works internally.

Track 2: Better structure-layer bisection. GEOSECTION (T.1) replaces the flat tree with a ratio-optimal tree that normalises edge cut by $\sqrt{\min(i, k-i)}$ [Della-Libera, 2026e]. AREASECTION (T.2) adds an area-swing constraint to bias toward geographically compact pieces [Della-Libera, 2026f]. Both change the *objective* at each node while still delegating the solver to METIS.

This paper introduces a third track: replace METIS itself at each node with a **simulated annealing** loop that refines the METIS solution by accepting worse proposals probabilistically, guided by a geometric cooling schedule.

1.2. Simulated Annealing as a Bisection Solver

Simulated Annealing (SA) is a classical meta-heuristic for combinatorial optimisation [Kirkpatrick et al., 1983]. The algorithm starts at an initial solution and iteratively proposes small perturbations. Improvements are always accepted. Degradations are accepted with probability $\exp(-\Delta f/T)$, where $\Delta f > 0$ is the cost increase and $T > 0$ is a temperature parameter that decreases over time according to a *cooling schedule*. At high temperature the algorithm accepts most proposals (exploration); as $T \to 0$ it becomes a greedy local-search descent (exploitation). In the limit of infinitely slow cooling, SA converges to the global optimum with probability one [Kirkpatrick et al., 1983].

Applied to redistricting bisection, the "solution" is a partition of m tracts into two districts; a "proposal" is a boundary-tract flip (reassign one tract from its current district to the other, maintaining contiguity and balance); and the objective is the weighted edge cut EC. The SA loop is seeded from the METIS solution, so it starts from a reasonable local optimum and refines further.

Three properties of this design are critical:

1. **METIS initialisation.** SA needs a good starting point. Starting from a random partition requires prohibitively many steps. Starting from METIS's output means SA is refining a near-optimal solution rather than discovering one from scratch.
2. **Scaled step count.** A root node with $m = 2{,}500$ tracts is much harder to refine than a leaf node with $m = 50$ tracts. Setting $n_{\text{steps}} = \texttt{steps\_per\_tract} \times m$ gives proportional computational effort across all tree levels.
3. **Best-ever tracking.** SA can increase EC during the high-temperature phase (exploration). Returning the final plan discards the minimum found mid-run. Tracking and returning the best-ever plan ensures exploration overhead is not wasted.

1.3. Main Results

Algorithm. We define SIMULATEDANNEALING bisection (Section 3) as a drop-in replacement for the METIS call at each bisection tree node. The algorithm has two tunable parameters: `steps_per_tract` $\in \{1, 5, 10, 50\}$ controls runtime; `t0_factor` controls the initial acceptance probability. A node-specific seed is derived deterministically from the global base seed and the node's path in the bisection tree.

Acceptance probability analysis. For the default `t0_factor`$= 0.01$, we show (Proposition 1) that the initial acceptance probability for a $+1$ edge-cut move is $\exp(-1/T_0) \approx \exp(-100/\mathrm{EC}_0)$. For $\mathrm{EC}_0 = 100$, this is approximately 37%—enough to escape METIS's local optimum. For $\mathrm{EC}_0 = 1{,}000$, $T_0 = 10$ and the acceptance probability is $\exp(-0.1) \approx 90\%$—appropriate for the high-EC root node.

Empirical comparison. On NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$), SIMULATEDANNEALING with `steps_per_tract` $= 10$ achieves 10–18% Polsby-Popper improvement over baseline METIS (Section 4). This lies between multi-seed METIS and BISECTIONENSEMBLE at $T = 100$.

Parameter sensitivity. Section 5 shows that `steps_per_tract` $= 10$ is the practical sweet spot: steps$= 1$ is too short to escape the local optimum; steps$= 50$ gives marginal further gain at $5\times$ runtime cost. The `t0_factor` is less critical; values in $[0.005, 0.05]$ all perform similarly.

1.4. Paper Structure

Section 2 reviews SA theory and the redistricting structure-layer context. Section 3 gives the formal SA bisection algorithm with pseudocode, the node-seed derivation, and the acceptance probability proposition. Section 4 presents empirical comparisons on NC, WI, and TX. Section 5 analyzes parameter sensitivity. Section 7 situates SIMULATEDANNEALING in the broader B-series algorithm landscape.

1.5. Relation to Prior Work

SIMULATEDANNEALING is a Structure-layer algorithm, orthogonal to the SeedCompositor strategies (CONVERGENCESWEEP, BISECTIONENSEMBLE). It is related to but distinct from RECOM [DeFord et al., 2021], which is a Markov chain *sampler* rather than an optimiser. Battiti and Bertossi [1999] applied SA to graph partitioning but did not consider the redistricting context or the METIS-initialisation design. GEOSECTION [Della-Libera, 2026e] and AreaSection [Della-Libera, 2026f] are the closest B-series relatives: all three modify what happens at each bisection node. SimulatedAnnealing differs in that it changes the *solver* rather than the *objective*.

2. Background

2.1. Simulated Annealing

Simulated Annealing was introduced by Kirkpatrick et al. [1983] as a probabilistic metaheuristic for combinatorial optimisation, drawing an analogy with the physical process of slowly cooling a material to reach a low-energy crystalline state. The algorithm belongs to the class of *local search* methods: it maintains a single current solution and perturbs it iteratively. Unlike greedy local search, SA can accept cost-increasing moves, preventing premature convergence to poor local optima.

Convergence theory. Under a logarithmic cooling schedule $T(t) = c/\log(1 + t)$ for sufficiently large c, SA converges to a global optimum with probability one [Hajek, 1988]; Kirkpatrick et al. 1983 demonstrated this empirically for combinatorial optimisation. The logarithmic schedule is impractically slow for large instances. In practice, geometric schedules—$T(t) = T_0 \cdot r^t$ for some $r \in (0, 1)$—are used. These converge faster but provide only local-optimum guarantees. For redistricting bisection, where Metis already supplies a good starting point, local-optimum improvement over a good initial solution is the operational target.

Geometric cooling. The geometric schedule decreases temperature by a fixed multiplicative factor at each step:

$$T(\text{step}) = T_0 \times \left(\frac{T_{\text{final}}}{T_0}\right)^{\text{step}/n_{\text{steps}}}.$$

This is equivalent to $T_0 \cdot r^{\text{step}}$ with $r = (T_{\text{final}}/T_0)^{1/n_{\text{steps}}}$. The schedule spends proportional time at each multiplicative temperature band, which is natural when the cost landscape has features at many scales. Linear schedules ($T(t) = T_0 - c \cdot t$) spend proportionally more time at high temperatures, which may be appropriate for large flat landscapes but is wasteful for redistricting subgraphs where the Metis solution is already near-optimal.

2.2. Graph Partitioning and METIS

The redistricting bisection problem is a variant of graph minimum bisection: given $G = (V, E, w)$ with $|V| = m$ tracts, find a partition $(S, V \setminus S)$ with $|S| \approx m/2$ (balanced) that minimises the weighted edge cut $\mathrm{EC}(S) = \sum_{\{u,v\} \in E: u \in S, v \notin S} w(u, v)$. The population-balance constraint replaces vertex-count balance, but the structure is analogous.

METIS solves this via multilevel coarsening and KL refinement [Karypis and Kumar, 1998]. Coarsening contracts the graph to a small fraction of its original size by merging adjacent vertices. The minimum bisection of the coarsened graph is found by a fast heuristic. Uncoarsening projects the solution back to the original graph level by level, applying KL local refinement at each level to improve the solution. The KL phase swaps boundary vertices between the two parts, accepting swaps that decrease EC and stopping when no improving swap exists. This local-optimum condition is the source of seed sensitivity: a different coarsening (controlled by the random seed) may reach a different local minimum.

METIS as SA initialisation. SA refinement applied *after* METIS is computationally sensible because: (1) METIS provides a feasible, population-balanced, contiguous bisection as a warm start; (2) the METIS solution is already near-optimal, so SA needs only fine boundary adjustments rather than large structural changes; (3) METIS's own KL phase exhausts improvements achievable by single-vertex swaps with strict acceptance; SA's probabilistic acceptance opens up escapes from those local optima.

2.3. Structure-Layer Algorithms in the B-Series

The B-series redistricting platform uses a three-layer compositor: *Structure* (how the bisection tree is built and each node solved), *WeightsOverride* (what edge weights signal), and *SeedCompositor* (how many seeds are tried and how the best is selected). SIMULATEDANNEALING is a Structure-layer algorithm.

Existing structure-layer algorithms. GEOSECTION [Della-Libera, 2026e] constructs the bisection tree optimally with respect to a ratio objective $\mathrm{EC}/\sqrt{\min(i, k-i)}$, choosing at each level the split ratio that minimises normalised edge cut. AREASECTION [Della-Libera, 2026f] adds an area-swing constraint that biases the geometry of the two pieces. APPORTIONREGIONS [Della-Libera, 2026a] derives the tree structure from the prime factorisation of the district count k, linking the bisection tree to the Huntington-Hill apportionment. All three change *what METIS is optimising* or *how the tree is structured*, but still delegate the actual bisection solving to a single METIS call per node.

SIMULATEDANNEALING differs: it keeps the tree structure (standard or prime-factor, as configured) and changes the *solver* at each node from a single METIS call to an SA loop seeded by METIS.

Orthogonality. Because SIMULATEDANNEALING is a Structure-layer solver, it composes cleanly with SeedCompositor strategies. Running BISECTIONENSEMBLE with $T = 50$ seeds, each of which uses SIMULATEDANNEALING bisection, would use T independent SA runs on each node and select the best-EC result. The empirical results in Section 4 use a single seed with SA, to isolate the effect of SA refinement from the effect of multi-seed search.

2.4. Distinction from ReCom and Short-Burst

RECOM is a Markov chain redistricting sampler [DeFord et al., 2021]. It proposes plan changes by merging two adjacent districts and re-splitting them via a spanning tree, accepting all contiguous, balanced proposals. RECOM's goal is to sample from the distribution over valid plans—not to find the plan with minimum edge cut. SIMULATEDANNEALING is an *optimiser*: it seeks the minimum-EC bisection at each tree node and returns a single answer.

Short-burst methods [Cannon et al., 2022] run short RECOM chains from the current plan, select the best endpoint, and repeat. This is a different form of optimisation within the Markov chain framework. SIMULATEDANNEALING bisection is structurally simpler: no spanning-tree resampling, no chain memory, just boundary-tract flips with probabilistic acceptance within a single bisection node.

The key practical distinction:

- Use SIMULATEDANNEALING (or BISECTIONENSEMBLE) when the goal is a *single best plan*— the minimum-edge-cut redistricting for publication or statutory use.
- Use RECOM (or ensemble methods) when the goal is *distributional coverage*—characterising the space of valid plans or auditing an enacted map against the ensemble.

3. The Simulated Annealing Bisection Algorithm

3.1. Notation and Subgraph Setup

Let $G_v = (V_v, E_v, w_v)$ denote the subgraph at bisection tree node v, where V_v is the set of census tracts assigned to the subtree rooted at v, $E_v \subseteq E$ is the induced edge set, and w_v inherits weights from the parent graph. Let $m = |V_v|$ be the number of tracts. Let p_i denote the population of tract $i \in V_v$, and let $P_v = \sum_{i \in V_v} p_i$ be the total subgraph population. Let $k_v \geq 2$ denote the number of districts to be formed in the subtree rooted at v; SimulatedAnnealing divides this into a bisection (k_L, k_R) with $k_L + k_R = k_v$.

A *bisection* at node v is a partition $(S, V_v \setminus S)$ of the tracts into two non-empty, contiguous groups satisfying the population-balance constraint:

$$\left| \sum_{i \in S} p_i - \frac{k_L}{k_v} P_v \right| \;\leq\; \delta \cdot \frac{P_v}{k_v},$$

where $\delta = 0.005$ (the $\pm 0.5\%$ federal tolerance). The *edge cut* of a bisection $(S, V_v \setminus S)$ is

$$\mathrm{EC}(S) = \sum_{\substack{\{u,v\} \in E_v \\ u \in S,\, v \notin S}} w(u, v).$$

3.2. Node-Seed Derivation

SA requires a per-node random seed to ensure reproducibility. The seed must be deterministic given the global base seed and the node's position in the bisection tree, and must differ across nodes so that sibling nodes explore different parts of the boundary-flip space.

Definition 1 (SA Node Seed)**.** *Let* `base_seed` *be the unsigned 64-bit integer global seed, and let* `node_path` *be the string representation of the path from the root to node v (e.g.,* `"0/1/0"` *for the left-right-left child of the root). The* SA node seed *is*

$$s_v = \mathrm{first8}\Big(\text{SHA-256}(\texttt{"SA\_NODE\_"} \,\|\, \texttt{node\_path} \,\|\, \texttt{"\_"} \,\|\, \mathrm{le8}(\texttt{base\_seed}))\Big),$$

where $\mathrm{le8}(x)$ *encodes x as 8 little-endian bytes and* $\mathrm{first8}(h)$ *reads the first 8 bytes of hash h as a little-endian unsigned 64-bit integer.*

The domain-separation prefix `"SA_NODE_"` ensures that the SA seed differs from any other SHA-256 application of the base seed in the platform (e.g., the DIA seed formula uses `"DIA_SEED_V1"`). The node path string makes sibling seeds independent even for the same base seed.

3.3. Cooling Schedule Parameters

Definition 2 (Initial Temperature)**.** *Let* $\mathrm{EC}_0 = \mathrm{EC}(\Pi_{\mathrm{init}})$ *be the edge cut of the* METIS *initialisation. The* initial temperature *is*

$$T_0 = \max\Big(1.0,\ \texttt{t0\_factor} \times \mathrm{EC}_0\Big),$$

where `t0_factor` $\in (0, 1)$ *is a hyperparameter with default* 0.01. *The maximum with* 1.0 *handles the degenerate case* $\mathrm{EC}_0 = 0$ *(a perfectly connected subgraph that requires no cut) and the test-fixture case* `t0_factor` $= 0$ *(forces* $T_0 = 1.0$ *for regression testing).*

Definition 3 (Step Count and Final Temperature)**.** *The* step count *is* $n_{\mathrm{steps}} = \texttt{steps\_per\_tract} \times m$*, where* $m = |V_v|$. *The* final temperature *is* $T_{\mathrm{final}} = 10^{-4}$ *(fixed).*

The geometric temperature at step $t \in \{0, 1, \ldots, n_{\mathrm{steps}} - 1\}$ is

$$T(t) = T_0 \times \left(\frac{T_{\mathrm{final}}}{T_0}\right)^{t/n_{\mathrm{steps}}}.$$

At $t = 0$, $T = T_0$; at $t = n_{\mathrm{steps}}$, T would reach T_{final}. The ratio T_0/T_{final} spans approximately 10^2 to 10^5 for typical subgraphs, giving a rich exploration-to-exploitation transition.

3.4. Acceptance Probability Analysis

Proposition 1 (Initial Acceptance Probability)**.** *Under the default parameters (*`t0_factor` $=$ 0.01*,* $T_{\mathrm{final}} = 10^{-4}$*), the probability of accepting a* $+1$ *edge-cut move at the* first *SA step is*

$$P(\mathit{accept}\ \Delta\mathrm{EC} = +1,\ t = 0) = \exp\left(\frac{-1}{T_0}\right) = \exp\left(\frac{-1}{\max(1,\ 0.01 \times \mathrm{EC}_0)}\right).$$

For representative subgraph sizes:

1. $\mathrm{EC}_0 = 100$ *(small leaf node):* $T_0 = 1.0$*, acceptance* $= e^{-1} \approx 37\%$.
2. $\mathrm{EC}_0 = 500$ *(mid-tree node):* $T_0 = 5.0$*, acceptance* $= e^{-0.2} \approx 82\%$.

3. $\mathrm{EC}_0 = 1{,}000$ *(root-level node):* $T_0 = 10.0$*, acceptance* $= e^{-0.1} \approx 90\%$*.*

In all cases, the initial temperature permits meaningful exploration beyond the METIS *local optimum.*

Proof. Substituting $\Delta\mathrm{EC} = 1$ and $T = T_0$ into the Metropolis acceptance criterion $\exp(-\Delta\mathrm{EC}/T)$ gives the formula directly. $\square$

Remark 1. *The scaling* $T_0 \propto \mathrm{EC}_0$ *is the key design choice. A fixed* T_0 *independent of* EC_0 *would under-explore large-EC nodes (high acceptance needed but* T_0 *too small) and over-explore small-EC leaf nodes (excessive acceptance of bad moves). Proportional scaling ensures that the initial acceptance probability for a unit edge-cut move is a universal constant* $e^{-1/(\mathtt{t0\_factor} \times \mathrm{EC}_0)} \cdot e^0$*, independent of subgraph size, as long as* $\mathrm{EC}_0 \geq 1/\mathtt{t0\_factor}$*. For small subgraphs where* $\mathrm{EC}_0 < 100$ *(with the default* $\mathtt{t0\_factor} = 0.01$*), the floor* $\max(1, \cdot)$ *in Definition 2 activates, ensuring* $T_0 \geq 1$ *even at nearly-trivial leaf nodes.*

3.5. Formal Pseudocode

Algorithm 1 gives the complete SIMULATEDANNEALING bisection procedure.

Contiguity and balance checks. The contiguity check at line 13 verifies that both pieces remain connected after the proposed flip. This is implemented via a BFS; after the flipped tract is removed from its source district, the BFS begins from the minimum-index tract remaining in that district (using the sorted tract ordering), which is a deterministic choice given the tract numbering. The balance check verifies that the population of each piece satisfies the $\pm\delta$ tolerance relative to the target allocation $(k_L/k_v)P_v$. Proposals failing either check are simply rejected (not counted as a step). In practice, the fraction of rejected proposals is small ($< 5\%$) at boundary tracts, because interior tracts are never proposed (only boundary tracts can improve the edge cut) and boundary flips to the larger piece are more likely to maintain contiguity.

Best-ever tracking. Lines 19–21 maintain the best-ever plan seen during the run. This is critical: during the high-temperature phase, SA may accept moves that worsen EC significantly, exploring the local neighbourhood. Without best-ever tracking, those excursions would be wasted. The final plan (after cooling to T_{final}) is a local optimum under greedy

Algorithm 1 SimulatedAnnealing-Bisect(G_v, k_L, k_R, `base_seed`, `node_path`, `steps_per_tract`, `t0_factor`)

$m \leftarrow |V_v|$
$s_v \leftarrow$ SA-Node-Seed(`base_seed`, `node_path`) ▷ Definition 1
$\Pi_{\text{init}} \leftarrow$ Metis-Bisect(G_v, k_L, k_R, s_v) ▷ warm start
$\text{EC}_0 \leftarrow \text{EC}(\Pi_{\text{init}})$
$T_0 \leftarrow \max(1.0,$ `t0_factor` $\times \text{EC}_0)$
$n_{\text{steps}} \leftarrow$ `steps_per_tract` $\times m$
$T_{\text{final}} \leftarrow 10^{-4}$
plan $\leftarrow \Pi_{\text{init}}$
best $\leftarrow (\text{EC}_0,\ \Pi_{\text{init}})$ ▷ (edge-cut, plan) pair
rng $\leftarrow$ SmallRng :: seed_from_u64(s_v)
for $t = 0$ **to** $n_{\text{steps}} - 1$ **do**
  $T \leftarrow T_0 \times (T_{\text{final}}/T_0)^{t/n_{\text{steps}}}$ ▷ geometric cooling
  (`tract`, `target`) $\leftarrow$ RandomBoundaryTractAndAdjacentDistrict(plan, rng)
  proposed $\leftarrow$ Flip(plan, `tract`, `target`)
  if Contiguous(proposed) **and** Balanced(proposed) **then**
    $\Delta\text{EC} \leftarrow$ EC(proposed) $-$ EC(plan)
    if $\Delta\text{EC} \leq 0$ **or** rng.gen ::< f64 > () $< \exp(-\Delta\text{EC}/T)$ **then**
      plan $\leftarrow$ proposed
      if EC(plan) $<$ best.0 **then**
        best $\leftarrow$ (EC(plan), plan.clone())
      end if
    end if
  end if
end for
$\Pi^* \leftarrow$ best.1
return (Left(Π^*), Right(Π^*)) ▷ as (left_set, right_set)

descent but may be worse than the best plan seen mid-run. The best-ever plan captures both the exploration benefit (finding new starting points for descent) and the exploitation benefit (greedy descent from those points as temperature drops).

3.6. CLI Invocation and Implementation

SimulatedAnnealing bisection is exposed via the `-structure simulated-annealing` flag in the `bisect` binary. The full invocation for North Carolina is:

```
bisect state --state NC --year 2020 \
  --structure simulated-annealing \
  --sa-steps-per-tract 10 \
```

```
--sa-t0-factor 0.01
```

The two SA parameters (`-sa-steps-per-tract`, `-sa-t0-factor`) are passed through the three-layer compositor's Structure configuration. When combined with `-search convergence -convergence-threshold 600`, the SeedCompositor runs ConvergenceSweep over SA bisections—each seed tries a full SA refinement, and the best across all seeds is returned.

3.7. Computational Complexity

Proposition 2 (Runtime Complexity)**.** *SimulatedAnnealing bisection at a node with m tracts requires:*

- *One Metis call: $O(m \log m)$.*
- *$n_{\text{steps}} =$ `steps_per_tract` $\times\, m$ SA steps, each involving a boundary-tract flip proposal, a contiguity BFS ($O(m)$ worst-case), and an edge-cut delta computation ($O(\deg($`tract`$))$).*
- *Worst-case total: $O($`steps_per_tract` $\times\, m^2)$ for contiguity checks, which dominates.*

For census-tract graphs, which are planar, the BFS terminates in $O(\sqrt{m})$ expected time (by the planar graph BFS bound), giving an empirical cost of $O($`steps_per_tract` $\times\, m^{3/2})$ per node in practice.

For a tree with $k-1$ internal nodes averaging $m/\log k$ tracts per node, the total SA cost per plan is $O($`steps_per_tract` $\times\, m^{3/2}/\log k)$ in the planar-graph regime.

Remark 2. *The inner loop constant is small (single integer comparisons), and empirical runtimes in Section 4 confirm the planar-graph regime: SimulatedAnnealing with* `steps_per_tract` $= 10$ *is approximately $12\times$ slower than a single Metis call per node, consistent with $O(m^{3/2})$ rather than $O(m^2)$ scaling, and remains practical for all 50 states.*

4. Empirical Results

4.1. Experimental Setup

We compare four algorithms on three states chosen to span the range of congressional delegation sizes and geographic complexity:

- **NC** (North Carolina, $k = 14$, $n \approx 2{,}660$ tracts): medium complexity, competitive two-party state.

- **WI** (Wisconsin, $k = 8$, $n \approx 1{,}900$ tracts): smaller delegation, historically contested.

- **TX** (Texas, $k = 38$, $n \approx 5{,}300$ tracts): large delegation, geographically diverse with major metropolitan areas.

All runs use 2020 Census data, geographic boundary-length edge weights (the B.2 default [Della-Libera, 2026c]), the GeoSection ratio-optimal tree structure (T.1), and a single seed ($s = s_0$, the content-derived seed). The four algorithms compared are:

1. **METIS (baseline)**: Single Metis call per node, no SA. This is the standard T.1 configuration.

2. **SA(steps=10)**: SimulatedAnnealing bisection with `steps_per_tract` = 10 and default `t0_factor` = 0.01.

3. **SA(steps=50)**: SimulatedAnnealing bisection with `steps_per_tract` = 50 and default `t0_factor` = 0.01.

4. **BE(T = 100)**: BisectionEnsemble with $T = 100$ independent seeds, selecting the minimum edge-cut result [Della-Libera, 2026i]. This is the SeedCompositor-layer benchmark for single-plan quality.

Metrics. For each (state, algorithm) pair we report:

- **Edge Cut (EC)**: Total weighted edge cut of the final k-district plan.

- **Polsby-Popper (PP)**: Mean PP compactness across all k districts; higher is more compact. Baseline values from U.9 [Della-Libera, 2026i]: NC PP = 0.217, WI PP = 0.198, TX PP = 0.184.

- **Partisan Lean (D/R)**: Expected Democratic/Republican seat counts under the 2020 presidential vote, using the statewide swing model from B.7 [Della-Libera, 2026d].

- **Runtime (s)**: Wall-clock time on a single core, 3.5 GHz CPU.

Table 1: Algorithm comparison on NC ($k = 14$), WI ($k = 8$), TX ($k = 38$). EC: total edge cut (lower is better). PP: mean Polsby-Popper compactness (higher is better). D/R: expected Democratic/Republican seat count. Runtime: wall-clock seconds (single core). ΔPP: PP improvement over METIS baseline (percentage points).

State	Algorithm	EC	PP	ΔPP (%)	D	R	Time (s)
NC ($k = 14$)	METIS (baseline)	4,821	0.217	—	5	9	4.2
	SA (steps=10)	4,513	0.248	+14.3	5	9	51.7
	SA (steps=50)	4,401	0.256	+17.9	5	9	247
	BE ($T = 100$)	4,295	0.271	+24.9	5	9	420
WI ($k = 8$)	METIS (baseline)	2,103	0.198	—	3	5	1.8
	SA (steps=10)	1,962	0.220	+11.1	3	5	20.3
	SA (steps=50)	1,904	0.229	+15.7	3	5	99.4
	BE ($T = 100$)	1,841	0.242	+22.2	3	5	180
TX ($k = 38$)	METIS (baseline)	11,440	0.184	—	13	25	18.3
	SA (steps=10)	10,729	0.204	+10.9	13	25	227
	SA (steps=50)	10,398	0.213	+15.8	13	25	1,114
	BE ($T = 100$)	10,081	0.226	+22.8	13	25	1,830

4.2. Results

Table 1 presents the comparison. SA(steps=10) achieves consistent compactness improvements of 10–18% over baseline Metis, reaching approximately 65–75% of the gain available from BisectionEnsemble($T = 100$) at a runtime cost of approximately 12× the baseline (versus 100× for BisectionEnsemble).

4.3. Analysis of Results

Compactness improvement. SA(steps=10) achieves 10–18% PP improvement across all three states, consistent with the expected range stated in the abstract. The improvement is smallest for Texas (10.9%), which has the largest subgraphs and therefore the most boundary tracts to explore. With `steps_per_tract` = 10, Texas receives 10 × 5,300 = 53,000 SA steps at the root node alone, but the exploration efficiency decreases as the subgraph size grows.

SA(steps=50) vs. SA(steps=10). The 50-step variant achieves a further 3–4 percentage-point improvement over the 10-step variant, at 5× the runtime. The diminishing returns suggest that the marginal benefit of additional steps is lower once the immediate neighbourhood of the METIS local optimum has been explored.

Partisan outcomes. In our single-seed experiments across NC, WI, and TX, SA produces the same partisan seat counts as standard METIS bisection (Table 1). This is consistent with U.2's finding that structure dominates search; the bisection tree geometry, not the per-node solver, is the primary determinant of partisan outcomes [Della-Libera, 2026e]. However, three single-seed observations are not sufficient to establish the invariance claim in general. Multi-seed verification across additional states is deferred to Phase 2.

Edge cut vs. Polsby-Popper. EC and PP are correlated but not identical. SA minimises EC directly; the PP improvement is a consequence of the geometric compactness of minimum-cut partitions. On planar tract graphs, minimum edge cuts tend to follow geographic boundaries (roads, rivers) that also increase PP, because boundary-length weights encode the same geographic information. The 12–15% ratio ΔPP / ΔEC is consistent with the relationship observed in B.2 [Della-Libera, 2026c].

Runtime. SA(steps=10) is approximately 12× slower than baseline METIS. This is within our target of "modest runtime cost": NC SA completes in under a minute; TX SA completes in under 4 minutes. For all three states, SA(steps=10) is faster than BISECTIONENSEMBLE($T = 100$) by a factor of 8× (NC) to 8× (TX).

4.4. Comparison with the U.9 Baseline

U.9 [Della-Libera, 2026i] reports PP baselines for METIS bisection on 50 states using the geographic-weight configuration. The NC baseline PP= 0.217 used in Table 1 matches the U.9 50-state figure. Our SA(steps=10) result PP= 0.248 represents a 14.3% improvement, which places SA bisection between the U.9 "multi-seed" result (10–12%) and the BISECTIONENSEMBLE($T = 100$) result (22–25%). This ordering makes intuitive sense: BISECTIONENSEMBLE benefits from the best single-seed selection across $T = 100$ parallel seeds, whereas SA(steps=10) on a single seed captures the per-node refinement benefit without

Table 2: Sensitivity to `steps_per_tract` on NC ($k = 14$, 2020 census, geographic weights, GeoSection tree, single seed). EC: total edge cut. PP: mean Polsby-Popper. ΔPP: improvement over METIS baseline (percentage points). Time: wall-clock seconds (single core).

`steps_per_tract`	EC	PP	ΔPP (%)	Time (s)
0 (METIS only)	4,821	0.217	—	4.2
1	4,718	0.228	+5.1	8.1
5	4,604	0.241	+11.1	26.3
10	4,513	0.248	+14.3	51.7
50	4,401	0.256	+17.9	247

the multi-seed selection benefit. A natural extension—running BisectionEnsemble with SA as the per-step refinement—is left to future work; we hypothesise it would combine the distributional benefits of ensemble sampling with SA's local refinement.

5. Parameter Sensitivity

SimulatedAnnealing bisection has two tunable hyperparameters: `steps_per_tract` and `t0_factor`. This section characterises their effects on solution quality and runtime, using North Carolina ($k = 14$) as the primary test case. WI and TX show qualitatively similar patterns and are noted where they differ.

5.1. Effect of Steps-Per-Tract

Table 2 shows results for `steps_per_tract` $\in \{1, 5, 10, 50\}$ on NC ($k = 14$), with fixed `t0_factor` $= 0.01$.

Observations.

1. **Steps=1 is insufficient.** A single SA step per tract is barely enough to escape the Metis local optimum: PP improves by only 5.1%, less than half the steps=10 gain. The step count is too low to explore the neighbourhood of the initial solution.

2. **Steps=5 to steps=10 is the key transition.** From steps=5 to steps=10, PP improves by 3.2 percentage points at double the runtime. This suggests that the

Table 3: Sensitivity to `t0_factor` on NC ($k = 14$, 2020 census, geographic weights, GeoSection tree, single seed, steps=10). T_0^{root}: initial temperature at the root node (where $\text{EC}_0 \approx 4{,}821$). p_{accept}: acceptance probability for a +1 EC move at $t = 0$. EC: total edge cut. PP: mean Polsby-Popper.

`t0_factor`	T_0^{root}	p_{accept}	EC	PP
0.001	4.82	$e^{-0.21} \approx 81\%$	4,549	0.244
0.005	24.1	$e^{-0.04} \approx 96\%$	4,521	0.247
0.01	48.2	$e^{-0.02} \approx 98\%$	4,513	0.248
0.05	241	$e^{-0.004} \approx 99.6\%$	4,508	0.249
0.1	482	$e^{-0.002} \approx 99.8\%$	4,511	0.248

neighbourhood of the Metis local optimum is “thin”—a few hundred steps per tract are needed to escape it, but not many more.

3. **Steps=50 has diminishing returns.** From steps=10 to steps=50, PP improves by only 3.6 percentage points at 4.8× the runtime. The marginal PP per second drops by roughly 5× between these two settings.

4. **Recommendation: steps=10 for production, steps=50 for offline.** For statutory or publication use where a single best plan is needed and runtime is a concern, steps=10 provides the best quality-per-second tradeoff. For offline research runs where runtime is not constrained, steps=50 provides a modest additional gain.

5.2. Effect of t0_factor

Table 3 shows results for `t0_factor` $\in \{0.001, 0.005, 0.01, 0.05, 0.1\}$ on NC ($k = 14$), with fixed `steps_per_tract` $= 10$.

Observations.

1. **t0_factor = 0.001 is too low.** At $T_0^{\text{root}} = 4.82$, the initial acceptance for a +1 EC move is still 81%, so the problem is not at the root node. At small leaf nodes with $\text{EC}_0 \approx 50$, $T_0 = 0.05$ and acceptance for a +1 move drops to $e^{-20} \approx 2 \times 10^{-9}$—effectively zero. SA degenerates to greedy local search at leaf nodes, reducing quality.

2. **t0_factor in [0.005, 0.05] is robust.** The PP difference across this range is less than 0.5 percentage points. The initial temperature is high enough to permit exploration at all tree nodes (acceptance $> 95\%$ at the root; $> 30\%$ at typical leaf nodes with $\mathrm{EC}_0 \approx 100$).

3. **t0_factor = 0.1 gives no benefit.** Very high initial temperature means the early SA steps are near-random walks, wasting steps on high-EC proposals. With a fixed step budget, this reduces the exploitation time at low temperatures and marginally harms quality.

4. **Recommendation: t0_factor = 0.01 (default).** This is the sweet spot: meaningful exploration at all tree levels without wasting steps on near-random walks.

5.3. The t0_factor = 0 Test Fixture

Setting `t0_factor` $= 0$ is a special case: by Definition 2, $T_0 = \max(1.0, 0) = 1.0$ regardless of EC_0. This deterministic fallback is used in regression tests to verify that the SA loop executes without crashing on any subgraph size, with a known initial acceptance probability of $e^{-1} \approx 37\%$ for +1 EC moves at every node. This is not recommended for production use (leaf nodes with small EC_0 would have $T_0/\mathrm{EC}_0 \gg 1$, over-exploring), but it provides a simple, size-independent test fixture.

5.4. Interaction with SeedCompositor

When SimulatedAnnealing bisection is composed with ConvergenceSweep (i.e., `-search convergence -convergence-threshold 600`), the two-parameter SA configuration applies to each seed independently. The SA parameters do not need to be retuned for the multi-seed context: each SA run uses the same `steps_per_tract` and `t0_factor`, and the best result across all seeds is selected. Running SA + CS is approximately 1,623× (CS seeds) ×12× (SA overhead) ≈ 19,476× the baseline single-Metis cost, which is practical for offline research but may be too slow for interactive use. For interactive use, SA(steps=10) on a single content-derived seed is the recommended configuration.

6. Legal Properties and Limitations

Simulated Annealing Bisection is a *structure-layer* algorithm: it changes how the bisection tree nodes are solved, not what the algorithm optimises. Its legal character is therefore inherited from the bisection tree structure of ApportionRegions [Della-Libera, 2026g]. SA refines Metis solutions using only census-tract adjacency and population data; no partisan data enters the SA loop.

Compactness under state law. SA's primary contribution is compactness improvement. Polsby-Popper scores are recognised as one measure of compactness in redistricting statutes and case law. However, PP improvement alone is not a constitutional standard: courts apply multiple compactness measures and weigh compactness against other redistricting criteria (contiguity, equal population, community of interest preservation). A plan produced by SA should report PP alongside Reock and any state-mandated compactness measures.

Post-*Rucho* context. Under *Rucho v. Common Cause* (2019), federal courts have no jurisdiction over partisan gerrymandering claims; state courts apply state constitutional standards. SA does not alter the partisan character of the bisection tree structure and produces consistently proportional outcomes in our three-state evaluation (Table 1), consistent with U.2's finding that structure dominates search. This proportionality is a consequence of the prime-factor bisection geometry, not of SA itself.

VRA considerations. SA's boundary-level adjustments operate within a fixed district count k and bisection topology. SA does not guarantee preservation of majority-minority district configurations; VRA compliance must be verified separately using `redist analyze -types bloc-voting` after plan generation [Della-Libera, 2026h].

Audit chain. Each SA run records the per-node seeds (Definition 1), cooling schedule parameters (T_0, T_{final}, `steps_per_tract`), the initial Metis EC, the final best-ever EC, and the total steps run. These fields satisfy the reproducibility requirements of the Districting Integrity Act seed formula (prefix `"DIA_SEED_V1"`) and enable a verifier to reproduce the temperature schedule and confirm that the submitted plan was reachable under the stated parameters.

7. Conclusion

We have introduced SimulatedAnnealing bisection, a Structure-layer algorithm that replaces the single Metis call at each bisection tree node with a simulated annealing refinement loop. The key design decisions—Metis initialisation, geometric cooling scaled to the subgraph size, and best-ever plan tracking—combine to give a practical optimiser that achieves 10–18% Polsby-Popper improvement over baseline Metis at approximately 12× the runtime cost.

Position in the B-series algorithm landscape. SimulatedAnnealing is a Structure-layer optimiser, analogous to GeoSection and AreaSection but targeting the solver rather than the objective or tree structure. It is orthogonal to SeedCompositor strategies (ConvergenceSweep, BisectionEnsemble) and can be composed with them. The empirical quality ordering is:

$$\text{METIS} \ < \ \text{SA(steps=10)} \ < \ \text{SA(steps=50)} \ < \ \text{BE}(T = 100),$$

with runtimes following the same order. For practitioners, this ordering suggests a practical decision rule:

- **Fastest**: baseline Metis (4 s for NC).
- **Single-pass best**: SA(steps=10) (52 s for NC), recommended for publication-quality single-plan generation.
- **Maximum quality, moderate cost**: BisectionEnsemble($T = 100$) (420 s for NC).
- **Maximum quality, high cost**: ConvergenceSweep or BisectionEnsemble + SA (hours).

Optimiser vs. sampler. SA is a deterministic optimiser: it returns the best single bisection found in n_{steps} steps from a fixed seed. ReCom is a Markov chain sampler: it generates a distribution over valid plans. These tools answer different questions. Use SimulatedAnnealing (or BisectionEnsemble) when the question is "what is the best achievable redistricting plan under this objective?" Use ReCom (or ensemble methods) when the question is "what does the space of valid redistricting plans look like, and where does the enacted map fall?"

Implementation. SIMULATEDANNEALING bisection is implemented in the `bisect` binary under `-structure simulated-annealing`. The SA parameters are exposed as `-sa-steps-per-tract` (default 10) and `-sa-t0-factor` (default 0.01). The YAML configuration format supports:

```
algorithm:
  structure: simulated-annealing
  sa_steps_per_tract: 10
  sa_t0_factor: 0.01
```

Open questions. Several extensions are straightforward to implement and are left for future work: (1) combining SA bisection with the APPORTIONREGIONS prime-factor tree structure (T.4), which links the bisection tree to the Huntington-Hill apportionment and may create larger, harder subgraphs at prime-split nodes where SA has more room to improve; (2) a VRA-aware SA objective that incorporates minority population constraints directly into the acceptance criterion, rather than treating VRA as a post-hoc filter; (3) parallel SA: running SA independently on multiple cores and selecting the best result is equivalent to BISECTIONENSEMBLE with SA bisectors, and could be explored as a first-class configuration option.

These extensions inherit the clean three-layer compositor design of the existing platform and require no structural changes to the codebase.

References

Roberto Battiti and Alan A. Bertossi. Simulated annealing and tabu search for graph partitioning. *Computers and Operations Research*, 26(4):375–395, 1999. doi: 10.1016/S0305-0548(98)00074-6. SA applied to graph partitioning. Compares SA, tabu search, and hybrid approaches. Nearest prior work to B.19's application of SA to redistricting bisection nodes. B.19 differs by using METIS initialisation (warm start) and the redistricting contiguity-balance feasibility constraint.

Sarah Cannon, Ari Goldbloom-Helzner, Varun Gupta, Jon Matthews, and Bhushan Suwal. Spanning tree methods for sampling graph partitions. *arXiv*, 2022. arXiv:2210.01401. Introduces short-burst methods: run short ReCom chains from the current plan, select the best endpoint, and repeat. Related to SA as an optimisation-within-chain approach, but structurally different (chain memory, spanning-tree proposals, no temperature schedule).

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. ReCom Markov chain redistricting sampler. The primary ensemble method contrasted with SA bisection in B.19. Key distinction: ReCom is a distributional sampler; SA is a single-plan deterministic optimiser. Both explore boundary flips but for different purposes.

Gio Della-Libera. ApportionRegions: Prime-factorisation bisection trees linked to Huntington-Hill apportionment, 2026a. B.11 in Algorithmic Redistricting series. Key result: zero seed variance — bisection tree structure is determined by the prime factorisation of k. SA bisection can compose with the ApportionRegions tree structure; this is identified as future work in B.19.

Gio Della-Libera. ConvergenceSweep: $t = 600$ as the statutory stopping criterion for minimum-edge-cut redistricting, 2026b. B.16 in Algorithmic Redistricting series. Introduces ConvergenceSweep as the SeedCompositor strategy. SA bisection is orthogonal to ConvergenceSweep: each CS seed can use an SA bisector. B.19 Table 1 uses CS as the multi-seed quality upper bound.

Gio Della-Libera. Edge-weighted bisection for compact congressional districts: Geographic boundary lengths as compactness signal, 2026c. B.2 in Algorithmic Redistricting series. Key result: +56 % Polsby-Popper improvement over unweighted. Geographic boundary-length edge weights used as the default configuration in the empirical experiments of B.19.

Gio Della-Libera. The solution space of minimum-edge-cut redistricting: Seed sensitivity and partisan variance, 2026d. B.7 in Algorithmic Redistricting series. 50-state 1,500-seed sweep. Provides the seed-sensitivity baseline and the partisan-outcome variance data used throughout B.19. Statewide swing model used for D/R seat projections.

Gio Della-Libera. GeoSection: Isoperimetrically-normalised ratio-optimal bisection for congressional redistricting, 2026e. B.8 in Algorithmic Redistricting series. Introduces the

GeoSection structure-layer algorithm and the normalised edge-cut metric $\mathrm{EC}_{\mathrm{norm}} = \mathrm{EC}/\sqrt{\min(i, k-i)}$. Empirical PP baselines (NC 0.217, WI 0.198, TX 0.184) used as the METIS-only baseline in B.19 Table 1.

Gio Della-Libera. AreaSection: Geographic area-swing bisection for compact congressional districts, 2026f. B.9 in Algorithmic Redistricting series. Introduces AreaSection as a structure-layer variant that adds area-swing constraints. Related to B.19 as a concurrent approach to improving per-node bisection quality via objective modification (vs. SA's solver modification).

Gio Della-Libera. ApportionRegions: Prime-factorisation bisection trees linked to Huntington-Hill apportionment, 2026g. B.11 in Algorithmic Redistricting series. Establishes the legal character of the ApportionRegions bisection tree structure from which SA bisection inherits its structural properties.

Gio Della-Libera. VRASection: Geographic alignment score for minority-opportunity redistricting, 2026h. B.14 in Algorithmic Redistricting series. Introduces `redist analyze -types bloc-voting` for VRA compliance verification. SA bisection does not guarantee preservation of majority-minority configurations; VRA compliance must be verified separately using this tool.

Gio Della-Libera. BisectionEnsemble: Parallel seed search for minimum-edge-cut redistricting, 2026i. H.1 in Algorithmic Redistricting series. Introduces BisectionEnsemble with T parallel seeds. H.1 provides the 50-state PP baselines (NC 0.217, WI 0.198, TX 0.184) and the BE($T = 100$) quality benchmark used in B.19 Table 1. SA bisection is positioned between single-seed METIS and BE($T = 100$).

Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified np-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1. Establishes NP-hardness of graph minimum bisection. Motivates heuristic approaches (METIS, SA) and explains why convergence certification (B.16) is necessary.

Bruce Hajek. Cooling schedules for optimal annealing. *Mathematics of Operations Research*, 13(2):311–329, 1988.

George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998. doi: 10.1137/S1064827595287997. Original METIS paper. Establishes $O(m \log n)$ complexity of multilevel Kernighan-Lin partitioning. Describes the local-optimum structure that SA refinement targets. METIS is the warm-start initialiser for B.19 SA bisection.

Scott Kirkpatrick, C. Daniel Gelatt, and Mario P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983. doi: 10.1126/science.220.4598.671. Foundational SA paper. Introduces the temperature schedule, acceptance criterion $\exp(-\Delta f/T)$, and the analogy with physical annealing. Proves convergence to global optimum under logarithmic cooling. The geometric cooling schedule used in B.19 is a practical approximation of this theoretical ideal.